Digital Transformation

Understanding Business Goals, Risks, Processes, and Decisions

Mathias Cöster, Mats Danielson, Love Ekenberg, Cecilia Gullberg, Gard Titlestad, Alf Westelius, and Gunnar Wettergren

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4. The Organisation of Digitisation

Ultimately, organising can be said to involve the division and coordination of work within and between organisations: the people, roles, departments, and/or external partners with which one will work, the skills each partner should have, and the mechanisms required to coordinate the work and ideas of each party so that they harmonise with the organisation's goals, strategy and business model.

There is a high degree of specialisation in certain contexts, such as in a fast-food restaurant or on the assembly line at a car manufacturer, where each person is responsible for a narrow work role: frying burgers, taking an order from a customer, or installing a tow hitch on a car. In others, such as a relatively new accounting firm with a few employees, the degree of specialisation is low and every employee may be responsible for performing accounting assignments for individual clients, working with sales, buying computers, and cleaning the office. Most organisations have something roughly called the "IT Department" and/or the "IT Manager". These kinds of designations say something about the prevailing attitude to IT in many organisations: that IT is a specialist area that is managed primarily by a specific department, just as sales are managed by the sales department and consolidation and analysis of financial performance are managed by the accounting department. This division of an organisation is usually based on the aim of achieving specialisation and becoming expert and efficient in a specific area.

There are also numerous tasks that are important to a business, but which the organisation may choose not to perform itself ("outsourcing"). Manufacturing companies have always faced the "make or buy" dilemma: Should car manufacturers make the parts for their cars, or buy them from

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other companies specialised in making each part? Should steelmakers buy iron ore and coal to produce their steel, or should they own their own mines? Services are also prime targets for such decisions; it is not unusual for reception, accounting, cleaning, and building maintenance to be managed by other companies. As mentioned in the introductory chapter, the trend towards XaaS (everything as standardised services for sale) means that product development, recruitment, and executive management can also be purchased or outsourced. This also applies to IT-related goods and services.

Regardless of how an organisation chooses to divide its work, the question of coordination remains. A relatively specialised role or department can impede a more general view of the organisation and the understanding of how one person's actions are intertwined with others', and how individuals contribute to the organisation's goals and strategies. Even if there is an understanding between roles and departments and overall objectives, there may still be dependencies that must be managed. The sales department, for example, is dependent upon the people who produce the goods or services sold by the organisation and on the people who analyse profitability, to name just a few. The parts, in other words, must be coordinated into a whole. Here again, there are various degrees: Should the parts be coordinated by allowing those highest up in the organisation to decide, or should employees slightly lower down also be allowed to make decisions? Should there be many and detailed rules, goals, and contracts for how activities should be performed, or will the organisation rely on employees and/or partners working in a manner that promotes the organisation's goals and strategies without excessive management? How can interaction between various parties be facilitated? Should the organisation expect it to happen by itself, or does it need to be encouraged?

As discussed in preceding chapters, digitisation is today a fundamental element of running a business or other organisation. There is much to indicate that digitisation issues have penetrated many different levels of organisations and have to some extent become "everyone's responsibility", or at least something that multiple employees can be expected to have ideas about. It is thus perhaps more difficult than ever to limit and allocate them to specific departments and/or roles. In the banking example provided in Chapter 2, the bank's IT department was responsible for only part of its IT deliveries, and some departments managed such issues independently. Customer patterns and other events inside and outside the organisation can now be tracked in different ways, and input information is often richer, available much more quickly, and to more employees than before. Communications between employees or between employees and outsiders are typically faster and accomplished through a wide range of digital channels. Increased servitisation has also made it easier to bring in digital resources from outside; software and related packages can now be bought as services, which has both advantages and drawbacks. As a result, the way organisations determine what should be done internally and what should be purchased is changing.

So, how should an organisation organise itself to realise the benefits of digitisation? How are issues related to digitisation allocated among different parties? How are the parties' needs and ideas related to digitisation coordinated so that they harmonise with goals, strategies, and business models? Who is allowed to present decision input and who is allowed to be involved and actually make decisions about implementation? Who will finance an initiative and who will be responsible for the final result? All of this is part of something called "IT governance". We have chosen to use the relatively everyday terms of "organising", "division of work" and "coordination" to describe this phenomenon. To paint a backdrop against which organising can be understood, the chapter begins with illustrations of a few issues related to digitisation that may arise in an organisation. Internal organisation will be covered next: who works with digitisation, various approaches to digitisation and the social aspects of coordinating business operations and IT. Finally, organising in partnership with external parties is addressed: what is outsourcing, the arguments for and against it, and how such relationships can be managed.

4.1. Background: A Few Questions surrounding Digitisation

Organisations have numerous questions about digitisation. Those we have chosen to highlight here are: 'Projects: Go or no-go?', 'Design of the project portfolio', 'The role of IT in goals, strategies, and business models', 'Balance between standardisation and freedom', and 'Level of reliability and security'.

Projects: Go or no-go? The implementation of specific systems, modules and applications entails numerous decisions. Should a new customer data system be deployed? Which supplier should be selected? What are the main benefits that such an investment should generate? For whom? What is the budget? What is the time horizon, i.e. for how many years can it be used? What implementation method should be used? Chapter 9 covers project decisions and management in greater detail.

Design of the project portfolio. There may be several parallel digitisation initiatives in an organisation, in which case the organisation will have to determine whether all of them should be run simultaneously or whether something needs to be postponed. This was illustrated in Chapter 2, where an IT manager at a bank described around seventy digitisation projects taking place simultaneously, and how continuous decisions as to whether all intended projects could in fact be started needed to be made (see also Chapter 8 on project portfolios). Ensuring that benefits from several projects at the same time are realised may be difficult, especially if those projects are large.

On the other hand, different initiatives may be related, meaning that the organisation will have to ask whether one project can generate benefits in another project. During an enterprise systems project carried out by BT, an international manufacturer of forklift and warehouse trucks, the common data management platform they implemented led (a few years later) to the company beginning to supply technicians who travelled to service customers' trucks with hand-held computers for tasks such as billing and registering the withdrawal of spare parts. This rather successful digitisation effort resulted in a reduction of back-office administration of truck service, an increased amount of service per service technician, better spare-part control, and increased status for the service technician role. The digitisation of service administration would not have been possible had BT not conducted the project on common-data-management platforms years earlier. It is not easy to predict what a project may lead to a few years down the line, but it is important to attempt to consider the long-term perspective.

The role of IT in goals, strategies, and business models: now and in the future. In addition to specific projects and any potential conflicts or synergies, there are more general questions about the role of IT in an organisation (as discussed in Chapter 3 on the digitisation of business models): How much in general should be invested in IT? Which parts of the organisation should be supported or developed via IT? How should this be accomplished? Does the organisation want to streamline and save money, or is there an opportunity to use IT more strategically? Is it necessary to generate new information about one part of the organisation? Can IT be used to enhance interaction with customers? With suppliers? What are the latest technological developments and how digitised are the competitors? Naturally, an organisation has to ask these questions when specific projects are discussed, but they can also arise in more general discussions. As noted, using IT to generate business benefit is not only a matter of supporting existing goals, strategies, and business models through specific projects; it also involves picking up on new ideas that may change the future direction of the organisation.

Balance between standardisation and freedom. Another question is whether the same IT solutions should be used throughout the organisation or whether there should be scope for local choices and adaptations, for instance in data definitions and standards. If different definitions of data are used by different departments of an organisation, it will make it harder to compare information between departments. Common standards and definitions, on the other hand, will reduce the flexibility of various departments, which may be important for meeting the myriad needs of, for instance, local markets. In the enterprise systems project at BT, mentioned above, the initial ambition was to achieve common definitions to increase transparency and comparability between the sales companies in various European countries. BT eventually had to modify this ambition because it met with tremendous resistance: the local companies were unwilling to change their definitions and put their operations on display. Numerous modifications were made so that the business system would be more closely aligned with local conditions. The end result was a common platform, but one that screened the local companies from each other.

Level of reliability and security. How reliable and secure digital resources should be is another question. As discussed earlier, IT plays different roles in different organisations. For some, like airlines and stock exchanges, IT is critical and must be immediately restored to service if there is a breakdown. That level of preparedness comes at a price, of course—a price that other organisations, where IT is not as business-critical, might not think is worth paying. Similar balancing can be seen in organisations that want to give customers and clients access to data, such as bank customers, patients in a regional health system, and users of tax agency services. There is no doubt that having access to their data via their own computers is valuable to customers and clients, but it also places an organisation at greater risk of hacking.

A high level of public access may be an obvious need that is not questioned—and not secured against unlikely risks. In connection with a widespread breakdown in 2016 in the data centre of a supplier operating the systems of many public organisations, the e-prescription function at the Swedish pharmacy chain Apoteket and the booking information system at Bilprovningen, the Swedish motor vehicle inspection service, were disabled. Some fifty business customers were affected by the breakdown. Those who had not paid for storage and operation at more than one geographical site were down for a week, and some lost business data. Questions, such as how customers and clients will be affected by an IT breakdown, and whether specific customer groups will be affected or a few specific parts of the organisation, must be discussed when the organisation is deciding the level of reliability and security. Reliability and security come at a cost.

4.2. Who Works with Digitisation?

As seen in the examples above, digitisation involves matters related to benefit, risk, flexibility, and transparency among other things—aspects that can be viewed differently depending on a person's place and role in the organisation and experience with, for example, past IT projects. People working in an IT department, which typically receives much of the blame for breakdowns, hacking, and data losses, may be more inclined to a high level of backup and security. They also tend to favour standardisation, since local deviations and quirks lead to additional work in purchasing, operations, maintenance, and user support. People who work directly with customers, on the other hand, are probably more inclined to give them access to various types of services and data and to push for customisation options and special solutions. Executives in the organisation probably think that uniform definitions of data are more important than middle managers do—the former want an overall picture that allows different parts of the organisation to be properly compared (see also Chapter 2 on how goals may vary between different levels and departments of an organisation).

One thing is clear: the question of who is involved in IT-related issues matters. Like other operational decisions, IT-related issues should be dealt with by a mix of people with an overall view of the organisation, a good understanding of the wider technical issues, detailed operational knowledge, detailed insight into technical matters, and good standing in the organisation. If decisions are to be accepted and complied with, the people involved in the decisions must have the respect of those employees who are most directly impacted. There is thus reason to think carefully about which departments and which levels will contribute their perspectives, where the decisions will be made, who will finance initiatives, and who will be responsible for them. There are, naturally, numerous differences in how IT issues are organised, depending on the type of IT decision (architecture versus decisions that will have a heavy impact on operations) and the type of organisation (with regard to size and strategy, for example). As organisations change and new IT issues arise, the organisation surrounding the IT issues may take on various forms, consciously or unconsciously, rapidly or cautiously. We do not believe that the organisation of IT is always necessarily the most rational for the specific situation. Consequently, we do not want to generalise to any great extent, but instead to present a few patterns that have been observed in studies, and provide a few concrete examples of the organisation surrounding IT issues in a few different businesses.

4.2.1. Who Decides?

According to some studies involving IT managers from various sectors and countries, the IT department is mainly responsible for decisions about IT architecture and IT infrastructure, although those decisions may be shaped by input from operations. This suggests that many managers believe that architecture and infrastructure are primarily technical issues and something they cannot or do not need to be involved in to any extent. In more clearly operations-related issues, it is more common for cross-functional groups to make the decisions, either unilaterally or taking into account views from individual departments. Operations-related decisions are made solely by IT managers in only a few organisations, which indicates an understanding that IT questions often involve the organisation itself, how it should be run, methods, allocation of roles, responsibilities, and so on. Apart from the austerity measures in the wake of the dot-com crash of 2000 and the financial crisis of 2008, when decisions about IT matters tended to be centralised around IT managers, the current trend is that more people are involved in IT-related decisions and local operations managers are more involved in strategic IT initiatives. The wider selection of IT suppliers, not least through the cloud, is sometimes cited as one important reason for this. Business managers describe cloud services as a way of liberating themselves from the IT department because cloud services are easy to buy and maintain (read more about cloud services in Section 4.5.1). It is perhaps not terribly surprising that operational employees have proven to be more positive about this increased decentralisation than IT employees (see also Chapter 5 on representing different parts of the organisation when making IT-related decisions).

When scholars have sought to identify successful approaches to organisation, they have looked at factors including how companies that are profit or growth leaders in their sectors have organised themselves around IT issues. Top financial performers tend to have more centralised IT decision-making. They may aspire to a more centralised IT environment in order to achieve synergies and cost efficiency; in that case, decision-making should also be centralised to a certain extent so that the IT environment does not grow into a jungle of disparate systems. Companies that have delivered the best growth tend instead to have more decentralised decision-making, where local units are given great latitude to initiate and run IT projects. The aim is to encourage innovation and alignment with external parties; if a company is to be able to rapidly respond to changes in customer preferences or the competitive landscape, it will not want to risk delaying critical decisions on IT investments because they must be filtered through a large group of representatives from various departments and levels of the organisation. This applies particularly to multinational organisations, where it is rarely reasonable to manage IT-related issues from a head office in another country when there are most likely skilled personnel and suppliers available on site.

These results should be interpreted circumspectly, as there are numerous other factors that can affect profitability and growth. It can be said, however, that these patterns are consistent with several studies that have shown how governance, whether or not it has to do with IT, should be designed according to a company's situation. Governance should be tighter, or looser, depending on how dynamic the environment is with regard to customer preferences, competitor actions, technological advances, the availability of substitute products, and political decisions. A more dynamic environment demands greater flexibility and independence among employees, usually characterised by a greater authority to make local decisions and greater scope to diverge from set budgets and goals (loose governance). In a more stable environment, success is not found to the same extent in rapid local response to external changes, but in delivering products efficiently. In such a situation, there is often more to be gained by coordinating the organisation, by means, for example, of more centralised decision-making, more detailed rules, and firmer demands to meet budgets and goals (tight governance). Adjusting governance mechanisms to the situation creates better conditions for the organisation to perform well.

Naturally, there is a greyscale between the two extremes of a "stable" environment and a "dynamic" one, and between "tight" and "loose" governance. An organisation aiming to achieve a cost-effective IT environment, and which has therefore centralised its IT decisions, also needs a structured approach to managing departures from this stance. A modification or investment that requires additions to the infrastructure that do not harmonise with the existing infrastructure might generate so much benefit that it is still considered worth implementing. There should thus be a process for evaluating such proposals. Accordingly, it is difficult to find any exact formula for how IT-related issues should be governed. Understanding that there are different types of situations and different types of governance and that these may harmonise with one another to a greater or lesser degree, is crucial.

4.2.2. Combining Perspectives from IT and Operations

According to a survey conducted in 2016 by the magazine *CIO Sweden*, about 60% of organisations have a formal IT advisory board that deals with IT-related issues on an ongoing basis. Just like formally established structures elsewhere, IT advisory boards seem to be most prevalent in larger organisations. The IT manager seems to have a relatively obvious role on these boards, as do business area directors in 86% of cases. The CFO is included a little over half the time, while the CEO is included in 44% of cases. The IT manager often presides over the board. A few illustrations will follow.

The multinational conglomerate Siemens has an IT advisory board at the corporate level and one for each business division. The boards at the division level include employees from both operations and IT, meet quarterly, and discuss which projects should be prioritised and how much money should be spent on IT, according to the needs of the division. Matters related to things like infrastructure and suppliers, however, are dealt with by the corporate-level board. Considering the size of Siemens and the breadth of its business, it is difficult to manage IT-related issues only at the central level.

Itab, a mail-order company that sells outdoor clothing via a number of subsidiaries in Europe, also has a combination of local and central decision-making, albeit not quite as formalised. The company wanted its subsidiaries to actively pursue IT issues, but they were not doing so, so the CIO now convenes meetings several times a year at each subsidiary, bringing together the local controller, head of accounting, managing director, and IT personnel. The subsidiaries vary in size, have varying needs when it comes to customer service, and there are numerous local systems, so the CIO therefore attempts to gather information about the needs of each subsidiary. They take certain matters further with the group CFO. In addition, they bring together representatives from all of the subsidiaries once a year to discuss matters related to infrastructure and integration.

The IT Advisory Board at Stokab, the dark-fibre provider owned by the City of Stockholm, Sweden, consists of the executive team, apart from the CEO, and meets about once a month. The various operations representatives are permitted to submit business that they want to be addressed at the meetings in advance, and the board usually also gives the various system owners a few minutes to talk about any ongoing changes. In general, the board discusses strategy, system administration, and system development and prepares input for upcoming investment decisions. The IT Advisory Board also has a budget to finance IT projects it has assessed as capable of generating value, and that need to be implemented quickly to respond to external changes. Without such a budget, there is a risk that projects will not be carried out because each operation is locked into its ordinary budget and thus does not have the budgetary space for more spontaneous projects.

The question of who should finance IT initiatives is not only a matter of flexibility in relation to emerging needs, but also concerns responsibility. The previously mentioned enterprise system project at the forklift truck manufacturer BT was centrally financed, and that caused difficulties in getting local units to accept responsibility for the project. The implementation of hand-held computers for service technicians a few years later was locally financed, but centrally monitored, and was characterised by stronger local responsibility. The degree of responsibility probably depends on several things, such as who was involved in initiating and deciding to carry out a project, but the source of financing is likely to play a major role.

In addition to advisory boards that regularly deal with IT-related issues, advisory boards, often called steering groups, are usually appointed for specific projects aimed at making decisions about supplier selection, the implementation method, and organisational changes that are within the framework of that specific project. These steering groups should also reflect a balance between operations and IT and between the central and local levels that aligns with the organisation's ambitions for IT utilisation.

Integration between IT and operations also takes place in a day-to-day context. The IT department at the Swedish Parks and Resorts entertainment group has recruited one person from the accounting department and another from marketing. They are meant to provide support for each function, so that the marketing department, for example, can manage much of the website itself. The weather report supplier AccuWeather has also loosened up the boundaries between IT and other operations in its day-to-day work. The company has an innovation team, for example, comprised mainly of former technicians from the IT department, who offer analyses to customers. Some of the tasks previously assigned to the sales department have been transferred to IT, because technical issues often arise in dealings with customers. For that reason, the technicians have also undergone training in listening and providing customer service by telephone. The Schools and Education Division of the City of Stockholm, Sweden, has created an

ICT department staffed by educators and IT personnel because the city wanted an environment in which multiple perspectives could contribute to further developing the organisation.

Even if IT has become more integrated with operations, and skills from various parts of the organisation are needed to understand how IT can be used, the CIO and the IT department still play a key coordinating role. Allowing the organisation to run IT projects any which way can be risky; different systems and cloud solutions need, at least sometimes, to be connected into some kind of whole. In addition, cost advantages in purchasing and maintenance should be possible if purchasing, at least to some extent, is made more centralised, provided that the standardisation does not complicate the work of users. As mentioned before, the point is to strike a balance between centralisation and decentralisation according to the organisation's situation. In a particular organisation, it might not matter that employees choose different ways of storing and sharing their documents, as long as it is done with a reasonable measure of care. Certain decisions, especially those related to IT architecture, are still often made by central IT departments, precisely because architecture is a matter that demands an overall approach. As an example of who works with digitisation, and how digitisation initiatives may develop over time depending on conditions in the organisation, we now present the case of a school that implemented an information system for planning and grading.

A primary school implemented an IT platform designed for planning and grading. The platform was initially perceived as burdensome and user-unfriendly. Planning had previously been done on computers or even written by hand, and shared with others at the individual teacher's discretion. Following the implementation of the new system, teachers were required to enter their planning, both short-term and long-term, into the system, according to pre-set menus. Assessment and grading were to be done in accordance with a matrix where different goals and criteria from the curriculum were stipulated. Planning and grading were now more automatically shared with colleagues, headmasters, pupils and their parents.

Not only did this require a new way of structuring planning and assessments, but it also became evident that additional measures, aside from the platform *per se*, were needed. When made visible to a larger audience, it became evident that formulations of plans and assessments varied a great deal among teachers, despite certain guidance from the pre-set menus. This raised discussions about the need to streamline formulations. Such discussions largely departed from the teachers' own experiences and wishes. Although headmasters organised meetings where teachers were meant to discuss these matters, teachers also organised their own meetings in smaller groups, often based on subjects or year-groups. Those teachers who already collaborated closely in their day-to-day work typically also gathered to discuss the meaning of certain criteria and content, and how planning and assessments could be formulated accordingly.

Another issue that arose during the implementation was the question of how planning and assessments were understood by the external audience, the pupils and parents. Teachers were of the opinion that parents in general did not bother to read lengthy formulations, and that formulations must not contain bureaucratic vocabulary that would obscure the basic message.

After about a year, many teachers were significantly more positive about the system. The shared storage of content between colleagues over the years was deemed to save a lot of time; the initial heavy workload had ultimately, it seemed, paid off. The system was also perceived to facilitate work, as it provided guidance, both through its features and the collegial discussions that had taken place over the year. Teachers now felt that the system helped them to obtain an overview and to ensure that they covered all parts of the curriculum when doing their planning and grading. Both teachers and headmasters furthermore believed that the discussions that had arisen in conjunction with the system's implementation were refreshing, as they directed attention to important pedagogical issues, such as how to interpret certain knowledge criteria and what to include in the planning of a certain theme in a certain grade. However, there was still variation in how teachers formulated their planning and assessments, and some teachers expressed a wish for clearer guidelines from headmasters.

We learn several things from the above case that need to be considered by those who are responsible for, or work with, digitisation in organisations:

- How the use of an IT tool unfolds depends on existing structures in the organisation, such as division of work and habits of collaborating.
- Involving those who are deeply involved in day-to-day work and who will be affected most by an IT tool enables the implementation to be more suited to their needs and ultimately more positively perceived.
- Documenting and storing data not only requires the appropriate technology but also comes with costs in the form of the users' time: time that not everyone may be willing to devote.
- Documenting and storing data therefore requires some sort of guidance and/or incentive. This could include agreed-upon standards, sanctions, rewards or, not least, internal motivation, whereby users see the value of documenting and storing data, for themselves as well as for others.

4.3. What Does Working with Digitisation Entail?

In addition to allocating joint responsibility for digitisation issues across central and local levels and between IT personnel and other employees, it may also be useful to think about the nature of digitisation issues and how an organisation wants to distribute and balance various types of issues. We previously mentioned architecture as a distinct category that should be managed by the people responsible for IT at the central level, but IT issues may also be differentiated in terms of routine work and innovative work. It is easy to get the impression that digitisation is all about innovation and creating change in an organisation, but as mentioned in Chapter 3, innovative approaches to using IT eventually become established approaches, and some organisations jump on trends much later than others. On the one hand, digitisation can thus be considered a potential tool for finding new ways of working and new products to offer, thus justifying expenditure of time and energy on it. On the other hand, existing digitisation needs to be taken care of, systems upgraded, and users supported, which means that people are required to work more regularly on IT-related issues. The first may involve projects that last a few months and must quickly generate value for the client, while the other may involve a temporal horizon of several years.

In IT contexts, this balance between different tasks has been called "bimodal IT" (a term reportedly coined by the research firm Gartner). In management research, the phenomenon has been noted more generally, and has been called "ambidexterity". Some would argue that it is more efficient to have employees specialise in one thing or another and that the innovative part of the IT department should in such cases work in the organisation to get closer to the needs of those outside the department (as discussed in the previous section). Others would argue that it is difficult to draw a hard and fast line between innovative and administrative employees, and that such a division risks creating an unfortunate alienation—perhaps of the administrative group above all. First, the two types of work are not entirely clear-cut; new ideas can improve the operation of existing systems and innovations are meant to eventually become part of day-to-day operations. Secondly, system expertise is spread more evenly among IT personnel if everyone works on both innovative and administrative aspects; otherwise, there is a risk that valuable knowledge essential to systems administration will be lost.

This brings us to the many different roles with links to digitisation that can currently be found in organisations. As mentioned in the previous section, the C-suite roles of CEO, CFO, and CIO are often members of an advisory board. It is perhaps unsurprising that the CEO is included, as they are ultimately responsible for the business. Nor does the presence of the CFO raise any eyebrows; accounting and finance functions have a long tradition of using IT to support transaction management, and to consolidate input. The chief marketing officer and the head of communications can also be key roles in digitisation; nowadays, identifying the opinions and preferences of customers and other stakeholders, or communicating messages to stakeholders often involves the use of a full palette of digital channels. There are actually no limits to the roles in an organisation that may be important in creating value through digitisation: it depends on the organisation's focus. The Chief Information Officer (CIO) is often emphasised as an important role. Most recently, one might have heard of roles such as Chief Digital Officer, Chief Data Officer, and Business Information Officer (BIO).

The latter is found, for example, on the staff of a Swedish bank, where the BIO is described as a sort of "sub-CIO". The CIO is a member of the corporate executive management team, while the BIO is a member of the executive team for their particular business area. The BIO is responsible for the business area's IT budget and reports to the business area manager. Depending on the organisation, the same title may cover fairly disparate tasks and areas of responsibility, and the various titles may also overlap. Consequently, we do not believe there to be any point in defining what all these titles entail. We can only say that digitisation has led to several of the more established roles having to manage such issues and to the emergence of new roles entirely, and we will leave it at that. Here, we will be focusing on the role of the CIO, which has been studied quite extensively and is often brought up when talking about the need to work bimodally.

Some researchers, such as Mark Chun and John Mooney, discuss the role of the CIO in two dimensions: the organisation's information systems (IS) strategy and information systems infrastructure. The first applies to the level of forward-thinking and risk-taking and the second to how systematic and coordinated the organisation's technologies and processes are. Based on this, four types of roles emerge in a CIO typology:

Triage Nurse/Firefighter. In an organisation with a risk-averse IS strategy combined with a fragmented infrastructure, the CIO's main focus is to keep costs down and make the best of existing technology. As the name suggests, the role involves prioritising urgent cases and putting out fires, rather than taking a long-term, big-picture approach. The role requires considerable technical expertise, as much of it has to do with fixing what does not work.

Landscape Cultivator. In an organisation with a risk-averse IS strategy but a more systematised infrastructure, the CIO focuses on the ongoing maintenance and integration of various applications and processes. The connection between IS and operations is more distinct than in the first role and the CIO thus needs to be skilled at tasks such as project management, change processes, and training. The emphasis, however, is on technical issues and the CIO does not intervene in strategy.

Opportunity Seeker. In an organisation with a risk-tolerant IS strategy and a non-systematised IS infrastructure, the CIO must work to identify gaps where processes can be improved with new technology.

Problem-solving and relationship-building are key skills in this role. There is thus an opportunity here to use IS to develop the organisation, but in the form of isolated initiatives rather than from an overall perspective.

Innovator and Creator. In an organisation with a risk-tolerant IS strategy and a systematised IS infrastructure, the CIO can concentrate on innovation and driving new revenues through new ways of using IS. Unlike the Opportunity Seeker role, this type of CIO is more engaged in the organisation's strategy and the role that IS should play in the organisation. Insight into the business strategy and the capacity to negotiate and build relationships are especially important in this role.

In Chun and Mooney's study, there were roughly the same number of respondents in each role, but many wanted to move toward the more innovative role. In simple terms, the CIO could be called either the "IT manager" who seeks to keep infrastructure alive and make sure the organisation gets as much benefit as possible for the money it invests, or the "Chief Innovation Officer", who is more visionary and tries to find new ways to generate revenue through IS organisation-wide. The latter typically had a business background rather than a technical background. Some scholars point to the general trend that it is becoming more common for the CIO to come from a business background. The opportunity for a CIO to move towards a more innovative role also seems to depend largely on whether the infrastructure is stable (in which case less energy needs to be spent on technical problems and on training employees and persuading them of the value of IS) and the extent of IS integration with the organisation's strategies and processes (for example, IS may be considered a more integrated part of the business in a data analysis firm than in a construction company). A more recent study of Swedish CIOs (Magnusson et al., 2019), claims that CIOs tend to take fewer risks, partly due to IT governance models, and that such a defensive stance will diminish the role of CIOs in the digitisation of the organisation.

It has long been said that the CIO role should act as a bridge between IS and business, but the CIO's tasks and place in the organisational hierarchy vary, as indicated above. Many would argue that the CIO naturally belongs on the executive team, but some still report to the CFO, for example. While some have received support from an assistant CIO in order to become more involved in the organisation's processes and strategies and less involved in technical details, others argue that the CIO role is going to become redundant or be reallocated elsewhere in the organisation as the utilisation of IS becomes more integrated. The point here is that there are many and partially overlapping labels for IS-related roles and that their content probably depends on both organisational conditions and interactions with others, and the personal characteristics of the individual in the role. Studies have specifically focused on the interaction between IT managers and business managers and have found it to be important to the development of common goals regarding how IT creates value in an organisation. This will be addressed in the next section.

4.4. Social Aspects of Coordinating IT Specialists and Operations

So far, this chapter has mainly discussed more structural aspects of the division of work and coordination of IT-related issues, such as roles, tasks, and representation in decision-making fora, planning processes, and project teams. As noted numerous times in management research, structure and planning form only one dimension of running a business. The everyday actions and attitudes of employees can also have a profound impact on which initiatives are prioritised and realised, and the results of those initiatives. Ultimately, people will fill the roles, take on the tasks and represent their organisational home, and it is therefore equally important to understand what knowledge, attitudes, and values people may be carrying and how those affect the relationship between IT units and wider operations. The following section addresses how employees from different groups can approach each other and create good relationships.

First and foremost, we must ask what characterises a good relationship between IT units and operations. Some observers, such as Blaize Reich and Izak Benbasat, argue that this is characterised in the short term by business managers and IT managers understanding and supporting the respective unit's plans and goal setting and, in the long term, by a shared vision about the way that digitisation will contribute to the business. Their study is based on interviews with managers from Canadian insurance organisations, where IT was assessed as having a strategic role (defined according to the size of the IT budget and the hierarchical distance between the IT manager and the CEO). To achieve this kind of relationship, it is important to consider at least four factors.

Shared IT knowledge. This requires certain knowledge to be shared between business managers and IT managers. Business managers, for example, should be up-to-date with new technology and have experience participating in, or leading, IT projects. Why should only IT managers attend technology conferences? IT managers should have experience as line managers and should understand the industry, as well as the history and current situation of various business units. It goes without saying that other top management should keep track of the organisation's business environment, its goals and strategies, and what makes it successful (as discussed in Chapter 2) and the IT manager should thus be no exception. In so doing, the respective managers will not only better understand how IT can create value, how the other managers contribute, and the challenges that they (not least the IT manager) are facing, but can also personally participate in and feel a sense of responsibility for their counterparts' work, instead of looking out only for the performance of their own unit. This applies regardless of whether digitisation is used to solve a problem, improve an existing process, or transform the business and create new types of revenue. Depending on the hierarchical structure of the organisation, it may of course also be relevant for people other than managers to have insight.

IT experience. This refers to an organisation's previous IT-related projects. If past projects were successful, business managers may see more potential and value in digitisation and consequently involve IT managers in planning to a greater extent. Unfortunately, this logic also works in the reverse: there is a risk that a history of failed IT projects will damage the confidence of business managers and their willingness to collaborate.

Communication. This encompasses several forms of interaction between business managers and IT managers, ranging from informal meetings and emails to more formally designed interactions, such as project teams, committees, and cross-functional roles. There is much to indicate that more frequent communication promotes mutual understanding. *How* you communicate matters, of course: IT personnel may have to tone down their technical jargon and try instead to express themselves in terms with which the other party is familiar, which may differ between the various business units.

Links between business and IT planning. This refers to whether activities in the IT units and business units are planned separately or jointly, and whether the IT side is planned subordinately to operations, or whether it affects business planning. If business and IT are planned in a more integrated way, it is more likely that the parties will understand each other. Meetings at which both IT-related and non-IT-related project proposals are evaluated, and where space is allowed for continuous dialogue about priorities in the organisation, are important to integrated business and IT planning. Certainly, such meetings can lead to conflicts, but some scholars argue that cross-functional discussions with broader participation are more important than tight governance of the planning process from the top down (unless the organisation has very clear and stable goals, in which case there may be advantages to a top-down planning process).

Chapter 8 delves deeper into the prioritisation of projects based on the benefit they can generate and how well they align with other projects in the organisation. In some organisations, the prevailing idea is that IT should be delivered with a clearly identified buyer and seller, i.e., the IT department delivers a customer data management system ordered by the marketing department. The transaction may be regulated by a contract. Could such a strict division lead to difficulties in integrating the activities of IT units and business operations as recommended above? Or could it provide more reasons for communication between the parties since preferences, specifications, and conditions must be made explicit? It is not clear-cut that one solution is better than the other to achieve good relationships between IT and business operations. The success of organisations is not only dependent on structures, but also on people's actions within those structures.

The above factors, interacting with each other, can promote or inhibit the emergence of common goals and visions. For example, an IT manager who has experience with both IT and business may be highly successful at implementing a project, which helps create trust in the IT department. Such trust may be critical to whether a manager is asked to participate in cross-functional contexts, such as an IT committee, or to act in integrative roles. This creates multiple networks and facilitates more frequent communication and more integrated planning. Under these circumstances, it is possible that business managers and IT managers will develop a body of shared knowledge. When an IT manager is very familiar with business plans and goals, the chances that other IT personnel will also become familiar with them increase. Some organisations insist that IT personnel should regularly visit operations and that everyone must work with user support, and this knowledge can promote even better collaboration in the next IT project. This is, of course, something of an ideal scenario; naturally, the absence of any of the factors mentioned above may lead to shortcomings in other factors and thus a lack of shared goals and visions. Shared knowledge seems particularly important, as it helps to bridge difficulties in the relationship between IT and operations which may have arisen during less successful IT projects.

In more concrete terms, this means that organisations intent on improving integration between IT and business operations should think about how they recruit, train, organise, and reward their employees, e.g. by encouraging participation in courses and conferences, physically placing people from different departments together (the amount of information that can be spread and exchanged in an open-plan office or on the same floor of the building should not be underestimated), letting IT people manage parts of the business and vice versa, rewarding breadth of experience, and carefully discussing recent IT projects to learn lessons from any setbacks.

These integration mechanisms may be equally pertinent between groups other than IT and business, and have over the years often been brought up in management research, perhaps so much so that they sound obvious today. Nevertheless, they have often proven difficult to implement. For example, it has been determined that the social mechanisms mentioned above mainly promote more short-term harmonisation between IT and operations, which is indicative of the difficulty of creating enduring attitudes in an organisation. It has also been established that organisations that demonstrate good short-term harmonisation may also have a fairly long history—of perhaps ten years—of communication, successful projects, and job rotation. One possible reason for these difficulties is that all forms of specialisation lead to a specific vocabulary and a specific approach. Very extensive and frequent job rotation may be necessary to prevent this from taking root, and the situation may differ in various parts of the world. A study of digitisation in Japan found that it was common for business managers to have worked in the IT department for a few years and for IT managers to be responsible for other functions, often finance or planning, as well. Employee turnover is another possible reason. Regardless of how accepting and integrated existing employees are, the same attitudes are not automatically transferred to new employees, and may need to be learned, depending on their previous experience.

Whether harmonisation between IT and business operations is always a good thing is also open to discussion. There has been some talk of productive friction, which may be needed when the business environment is changing rapidly. If different parts of an organisation do not thoroughly understand each other's work and share goals and visions about how IT should be able to promote the health of the business, there is a risk that the organisation will be unable to respond to change as fast as necessary due to excessively one-sided perspectives and contentment with the status quo.

A case about e-government in municipalities is included below as an example of social aspects of coordinating IT specialists and operations. E-government—IT-supported initiatives to develop public operations, such as through the provision of services and information to citizens—has been developed widely in Sweden during the last decade. There are various forms of e-government: examples include new channels for communication between a municipality and its citizens, such as Facebook pages, chat forums, and YouTube clips, as a supplement to printed material and personal meetings, apps for reporting things that need to be repaired or addressed in different places, and portals where citizens can log in and obtain an overview of school, childcare, leisure activities and the like, all in one place, for example, when choosing a school. But what is the essence of e-government, and is it primarily a matter of technology or operations? Based on Gabriella Jansson's study of e-government, the case below describes two municipalities, A and B, both considered pioneers in e-government. Both municipalities have had a stable political government and clear operational goals for a long time.

Municipality A is characterised by far-reaching marketisation where services such as schools, eldercare and leisure activities are offered by both private and municipal actors, based on individual preferences. Municipality B has a long tradition of so-called citizen offices; places where citizens can obtain information and speak with both officials and politicians. This has been considered important in light of the large proportion of citizens with limited knowledge of the Swedish language.

The politicians in Municipality A have a clear vision of how IT should be used to generate benefits in the municipality, both for citizens and for employees. An example of the link between IT and political visions is the replacement of the previous IT portal, to which only municipal providers of welfare services had access, as it was considered important that it also be available to private actors. IT has thus been a means of supporting prevailing values. Some projects have been implemented quickly, with a lack of support from, for example, school principals as a result, but the advantage highlighted by officials is that the goals and vision of the IT projects have been marked by great clarity, which has helped to guide and legitimise the projects. The municipality has no separate strategy for IT, but instead takes as its point of departure the operations and what it wants to achieve.

In Municipality B, it is to a large extent the officials who have driven the development of e-government. The role of the politicians has mainly been to approve the officials' proposals for IT projects and to set budgetary frames. Costs and investments related to IT are also managed within the budget of each entity, which tends to reinforce fragmentation and short-term orientation. Projects have been added one at a time, leading to the feeling that no one has a complete grip on the various projects and how to prioritise them when financial resources are insufficient. There is an IT strategy, but it is perceived by some as too technically oriented. There is also the perception that IT projects are run by the IT department, despite an opinion among officials that IT is an organisational matter rather than a technical one. Even though the politicians are considered to be very committed and visionary with regard to the citizen offices, they seem to view e-government as a technical initiative rather than something that interacts with operations to any large extent.

The case above illustrates a couple of things to consider regarding coordination between IT specialists and operations:

- Depending on which actors are involved in IT-related decisions, those decisions will enjoy varying degrees of legitimacy and be more (or less) clearly related to operations.
- Previous IT projects can influence the attitudes of the different actors.

4.5. Outsourcing and Partnership with Suppliers

Our discussion of who participates in an organisation's digitisation, and how, and how various initiatives are coordinated, continues in this section, but the focus will now shift to situations of "outsourcing", in which digitisation is managed by external actors. The section begins with an account of what outsourcing entails and the possible reasons for and against outsourcing an organisation's digitisation. This is followed by a discussion about how outsourced operations are managed to align with the needs of the organisation.

4.5.1. What Does the Outsourcing of IT Entail?

Narrowly defined, outsourcing entails transferring something the organisation previously did for itself to an external supplier. The actual purchasing of such a service is merely sourcing, and could take place when a business is started or when a business switches from one supplier to another. In ordinary parlance, however, these have come to be known as outsourced services, rather than just the step from internally managed operations to contracting the job to an external party. It is in this wider sense that we address the phenomenon of outsourcing in this section.

IT support can involve both goods and services. Back when computers were still large and costly, companies commonly rented hardware rather than buying it themselves. After the breakthrough of the PC, it became more common for an organisation's computing power to come from PCs that it owned. At that point, outsourcing began instead to involve systems integration, networks, and telecommunications, the use of temporary contract workers instead of hiring people with the required expertise, or contracting for transmission capacity instead of trying to build and own the lines for transmission to the target audience. Since then, the pendulum has swung back and forth between owning and leasing computer capacity and between providing services of various kinds internally or buying them externally. Today, outsourcing is common in numerous areas of business, from computing capacity to systems development and operation, and even tasks that utilise computerised systems.

According to a 2019 US survey, application development is the most commonly outsourced service, and some of the fastest growing areas for outsourcing are network operations and IT security. Help-desk support and disaster recovery are particularly popular outsourcing areas for organisations intending to reduce costs, but as we shall see in Section 4.5.2, there may be several other reasons for choosing to outsource. Although it does occur, the outsourcing of entire processes is unusual. An example of outsourcing an entire process would be when Sweden Post redesigned its entire office network and, instead of only running its own post offices, began to partner with Swedish retail stores, such as ICA and Pressbyrån. For this to be possible, Sweden Post needed new systems that could be used by retail companies, whose employee turnover was high, and where employees were given only minimal training in handling mail and using systems. They also needed to install relevant systems in the partner stores and provide some training for store employees. Contracts were also required to specify what services the stores would perform and on what terms. In addition, Sweden Post chose to outsource much of the development and implementation work to consultant firms.

Markets for IT-related services are still developing. One prediction, based on a 2016 study carried out in the Nordic countries by Whitelane Research in partnership with PA Consulting, is that outsourcing is set to grow, especially in the areas of application development, maintenance, and testing. This scenario is not limited to the futures of large IT departments. As more goods and services are digitised, product and business development will increasingly become a matter of software and hardware development. As the total scope of such activities expands, companies naturally take advantage of the potential for specialisation and the development of markets for specialised services. Who does what may change over time, and going forward, some organisations will probably choose to conduct jobs that they previously outsourced internally, but, in general, the growth in outsourcing IT-related services is expected to continue.

Before we take a closer look at the reasons for outsourcing and how it can be managed, we will discuss a few types of outsourcing that have demonstrated particular growth in recent years.

Offshoring. One tendency is for companies to outsource certain operations abroad, for instance to India and the Baltic countries. This is usually called offshoring or nearshoring, depending on how far away the supplier is. This approach is commonly driven by cost-cutting, and it thus makes sense that the focus is on countries with a lower cost profile. According to a 2015 survey conducted by the magazine *CIO Sweden*, about 60% of survey respondents had outsourced part of their IT operations abroad, primarily to India, but also to the Baltic countries and the Czech Republic. Internationally, rather than being replaced by IT support, customer service, telephone sales, and other human support services are increasingly becoming the objects of offshoring. Language is important in interactions with customers, and Filipino companies are now competing successfully with Indian companies to become outsourcing partners for English-language services. Nearshoring and offshoring do not always result in cost reductions and efficiency. Language barriers, cultural differences, and differences in time zones can make it more difficult to make these partnerships work than it would be if they were outsourced within the same country or indeed if the services were performed internally.

Captive. Governance and coordination problems may cause an organisation to outsource IT-related operations abroad while still retaining control over them. This is known as a “captive solution”. In a narrow sense, this is thus not a matter of outsourcing, but it is still distinct from having an IT or customer service department within the same building as other departments, which is why we mention it here. Companies including Danske Bank, AstraZeneca, and Volkswagen have taken this route. After having engaged an Indian IT company for a few years, Danske Bank decided in 2014 to establish its own IT centre in Bangalore, India, and have them handle everything from major development projects to minor services of a simpler nature. EF, a company providing language-learning travel and exchange, has made a similar move, from previously having engaged an Indian consultancy to creating its own centre in Bangalore, the city sometimes called the "Silicon Valley of India".

Cloud services. "Cloud services" are another form of IT outsourcing that has grown in recent years and is expected to continue to do so. Cloud services involve the delivery of IT-related services over the Internet—which is sometimes depicted as a cloud in illustrations—instead of their being physically installed or performed at the purchasing organisation. Subscription is the most common form of contract, whereby the subscriber has access to the cloud for a finite period and often the option to adjust capacity upwards or downwards as needed. It has been possible to buy various forms of service from service agencies ever since the use of computers began in the twentieth century, but with the development of the Internet and the advent of increasingly faster and cheaper broadband, market opportunities are constantly expanding. Terms such as IaaS (Infrastructure-as-a-Service, i.e., the provision of storage space and computing power as a service), PaaS (Platform-as-a-Service, i.e., the provision of a platform for systems development), and SaaS (Software-as-a-Service, i.e., the provision of use of an online program) have been coined to describe the various types of standardised services available over the Internet. As the types of services—and their acronyms—multiplied, the all-encompassing XaaS (Everything-as-a-Service) was coined. These days, we are used to being able to get hold of everything under the sun as standardised services via the Internet, even as consumers. This might involve simpler services, such as document storage, email, and videoconferencing, but also survey instruments, business systems, databases, tools for accounting, billing, payment monitoring, etc. To some extent, programs from established software vendors are moving into the cloud. Office 365, the Microsoft Office package with an email program, word processing, document storage and sharing, is one example. Consumers have been using this type of IT via the Internet for a long time and organisations are increasingly choosing to manage these services through the Microsoft cloud, for example, instead of through their own infrastructure. There are also firms in other industries that have seen business opportunities in offering cloud services via their own extensive infrastructure. Amazon is now a leading supplier of cloud services and is challenging traditional vendors like Oracle and IBM. The traditional vendors are also taking action: IBM, for example, has sold parts of its hardware operations to concentrate more on cloud services.

Ultimately, outsourcing—in any form—can be said to involve striking a balance between control and flexibility. It can free up resources and thus enable more flexible action, but also entails fewer avenues through which to control how the specific aspect of operations is performed and how it fits into the rest of the organisation. Instead of direct management, it becomes a contractual matter. It is common to try and achieve clarity through specifications or service level agreements (SLA), in which the content and agreed quality of the service delivery are set out. This type of clear specification works better for a familiar, standardised service than for one that is flexible and changeable in terms of the access agreed between parties, response times, and so on, but things start to get tricky even with a helpdesk. What types of issues should the buyer expect to get help with? When is an issue actually resolved? What is the quality of the help? How should support for new products be dealt with? Is a new contract necessary or can the parties identify procedures for how soon the helpdesk must be able to provide (efficient and competent) support?

As we discussed in the introductory chapter, an organisation should be able to essentially build a Lego structure from purchased (cloud) services, integrated via systems that are either shared, or at least communicate with each other. In reality, however, standardisation is not quite that seamless, and the interface is not as clear and simple as the stacking of Lego bricks. The shape of the bricks and how they are combined may require both continuous contact and discussion. What kind of feedback do we have about the service execution and its suitability? What needs for development and adaptation exist now or will arise over time, and how should they be managed? What assurances does the organisation require before leaving a current and familiar service provider for a new one that seems promising but is as yet untried?

We will return to matters of governance and governability, but will first take a closer look at the reasons for and against outsourcing. Some of these are listed in Table 4.1 and will be further discussed in the following section.

Table 4.1 Reasons for and against outsourcing an organisation's IT.

Reasons for outsourcing	**Reasons against outsourcing**
Cost efficiency improvement	Cost of maintaining insight and monitoring
Focus on core business	Loss of skills
Flexibility	Data exposure
Development and improvement	

4.5.2. Reasons for Outsourcing

Cost efficiency improvement. As said, reducing costs is a key reason for outsourcing. A number of studies indicate that organisations that choose to outsource large parts of their IT management are often struggling with falling profits, high operating costs, high debt, and low liquidity. Studies that more explicitly studied reasons for outsourcing also clearly show that the potential to control and reduce costs is highly significant. Outsourcing can certainly contribute to lower costs; a supplier that has specialised in providing large-scale IT support can potentially deliver cost advantages (but also quality advantages, more on which later in this section) compared with providing support internally. When the airline company Scandinavian Airlines (SAS) decided in 2015 to weed its fragmented system flora and hand over a large part of IT operations to external partners, the airline's choice was an Indian firm, Tata Consulting. The CIO at SAS expressed the hope that costs in particular would be reduced, through several changes including shrinking the company's own IT department and digitising additional processes.

The cost argument is even more prominent in connection with the choice to outsource IT operations abroad, as there is a huge gap in wage levels between, for example, India and European countries. The choice to buy IT as cloud services is also usually characterised by cost savings arguments. In 2015, the Scandinavian grocery chain ICA moved parts of its IT operations to the cloud. After less than a year, they estimated that they had recouped the investment thanks to lower costs of administering databases and application servers. When the National Property Board of Sweden chose Microsoft Office 365 instead of continuing to run Office (including email) internally, the agency estimated that it would be about 40% cheaper. A government inquiry was conducted in 2015 that determined that Swedish government agencies generally needed to take better advantage of the opportunities provided by cloud solutions. The committee of inquiry estimated that large cost savings were possible, in part because the buyer of a cloud service can share operational personnel, software, and hardware with others instead of buying resources dimensioned to handle peak loads separately. When the usage of users in different organisations is aggregated, the peak loads of some will coincide with the off-peak usage of others, so that the combined capacity requirement will be much lower than the sum of everyone's peak loads.

Naturally, the savings are greater for organisations requiring either investment or new hires than they are for those that have already made the investments and hired the employees, because once resources have been spent, they can only be recovered to a certain extent. Outsourcing can thus be particularly economical for new organisations that have not yet had time to build up their own IT environments, customer service, accounting units, or whatever they are considering buying as a cloud service. Buyers of outsourced services often pay only for the capacity and storage space they actually use as opposed to paying for a peak load capacity that goes largely unused, which is typical for in-house operations. In turn, the supplier achieves economies of scale through the sharing of resources among customers. Upgrades are processed more often, but in smaller stages, which is thought to save the buyer a great deal of effort. It should be noted, however, that cloud services are not necessarily cheaper; after cost/benefit analysis, the location data company Foursquare concluded that it was more economical for them to rent space in a data centre and use their own hardware than to use a cloud-based database. If an organisation has highly specific systems developed in-house that still serve their purpose and an operating environment that is tried, tested, and efficient, the transition to cloud services, which are standardised to fit many, may require compromises, changes in working methods, or new learning and relearning, all of which make continued internal operations seem preferable.

The coordination gains that can be obtained through cloud services can instead be considered a special case of outsourcing. In traditional outsourcing, many of the potential cost advantages are based on efficiency gains related to the sharing of resources, such as premises, equipment, software, employees, and knowledge production. Although there is potential in many cases to improve cost efficiency through outsourcing, the underlying reasons are increasingly more strategic. We will now shift our focus to these.

Focus on core business. One more strategic reason for outsourcing is the opportunity to free up resources to focus on core business. It is a common view that IT which is not critical to the delivery of the value proposition can be outsourced. Email, intranet, accounting systems, and electronic billing are all examples of IT applications that fulfil a vital function in the day-to-day work of many, but which are established tools that do not in themselves constitute a way of further developing the unique selling points of an organisation's value proposition. That which constitutes the core of the value proposition varies. For a trading company, the logistics of delivering goods to customers can be a key component and its IT support must then be seen as strategic. In a high-tech firm, the main concern is instead the design and development of new products, in which case such IT support is of greater strategic significance. It is increasingly common for digital elements to be central product features: the logistics company's route planning, the battery charger manufacturer's charge management, the billing company's credit rating algorithm, the heat pump company's temperature control, or the car manufacturer's automated driving system. There may be a good reason to maintain control over such central components, which are strategically important to the business, by developing and managing them in-house or possibly in collaboration with a trusted partner.

The non-profit organisation NACE International argued in the above terms for a focus on core business when they chose to outsource their maintenance of infrastructure, as well as their cybersecurity activities, while keeping the development of their customer-facing applications in-house. The choice was based on the idea that back-office activities should be outsourced and that activities related to customers (i.e. activities that constitute the core business) should be conducted within the organisation. A Northern European social insurance agency cited the same reasons when they decided in 2015 to outsource systems development and certain parts of systems administration. The truck and bus manufacturer Scania has both outsourced and insourced system development and operation based on different assessments of the advantages and drawbacks of the respective alternative. British

Petroleum (BP) concentrated on forging alliances with other companies in the energy industry to encourage the growth of independent standard-systems developers so that it became possible to outsource systems development and operation to external parties. Different organisations come to different conclusions and the same organisation may change its position over time, so it is impossible to say in general that it is always wise to outsource one part of the chain.

Flexibility. The previously mentioned approach of varying usage according to the organisation's changing needs is another strategic reason for outsourcing. In 2015, the Swedish Transport Agency entered into a contract with IBM for the management of their data centres, worth nearly SEK 1 billion (EUR 100 million) over five years. The plan was to have IBM initially operate the data centres and eventually take them over entirely. As the need for a large data centre will vary, an ability to scale either up or down according to immediate requirements was considered important. Dimensioning to ensure the capacity to manage every conceivable peak load would be a waste of resources if the potential variations in needs are large and a cost-effective solution can be achieved through outsourcing. The Swedish Council for Higher Education knows there is a huge peak load a couple of times a year when eager prospective students who have just taken the national university aptitude test want to see the correct answers. This can be easily managed by outsourcing operations to a specialised supplier. There is even more reason to choose outsourcing instead of internal management if the type of resource requirement also varies in a way that cannot be satisfied by the organisation's own multi-skilled employees or other internal resources.

Cloud solutions, which are sometimes described as "IT on tap", usually entail tremendous flexibility. The European hotel chain Scandic started a project in 2015 to put its IT operations related to the website and the market into the cloud. A primary reason for this was the ability to scale up and down as needed, for example, when there was heavy traffic on the website in connection with events. The push for flexibility is not limited to existing IT use, and also applies to the development of new forms of IT use. Scandic wanted to offer cutting-edge solutions to their customers, and they concluded that they would not be able to develop tools fast enough on their own if they wanted to get a jump on the next big thing, for example within social media. Because the payment model for cloud solutions is typically tied to use of existing services—unlike purchased software or in-house/purchased development, where full payment is required before the company can begin to use it, and before it knows whether and to what extent it will be used—cloud services provide greater opportunities both to try competing services and (as long as the term of the cloud services contract is not too long) to switch suppliers. There is a drawback in flexibility, however, as often only standard solutions are on offer.

Development and improvement. A related strategic justification for outsourcing is the opportunity to leverage external skills and technology to learn new things and improve internal processes. This may apply to both strategic and less strategic aspects of a business. Although the argument above—that outsourcing should only be considered for things that do not make a critical contribution to the value proposition—is common, there are those who choose to outsource strategically important aspects precisely to gain access to high-level, up-to-date skills that can promote new ways of working. According to Deloitte's 2018 global outsourcing survey, one third of respondents were in fact willing to pay more for cloud services, if those services contributed to improved performance and innovative capabilities. Similarly, the choice to have outsourced activities managed abroad through either an offshoring or captive solution is not necessarily a matter of cost-cutting alone, but may also be made with a view to gaining access to a large number of highly educated software developers. The CIO at EF, a global language education company, has praised the skilled and dedicated personnel at the IT centre in Bangalore, who actively contribute suggestions for how operations can be developed. But it is by no means a given that access to high-level skills will lead to successful development. A manufacturer of smart electricity meters decided after a while to reverse the development of the function logic in the meters from a collaboration with a highly specialised development consultancy because the process of developing new functionality required more continuous dialogue and collaboration than the consultancy's contract model allowed for.

Thus, there may be several reasons behind outsourcing. Consultant reports in recent years have indicated that organisations in general are interested in outsourcing more of their IT, and that they increasingly regard outsourcing as a potential source of business development and innovation. Many argue, however, from both the buyer's and the supplier's side, that business development and innovation are not given much scope in connection with the outsourcing of ongoing operations. One reason for this may be a strong focus on costs in the contract and in ongoing management (more about this later in the chapter). One reason might be that it is more difficult to get feedback on needs and opportunities for development from a partner than from the organisation's own employees. First, it is thought that provisions concerning such feedback, for example from an outsourced helpdesk or customer service, need to be expressed in the contract. Secondly, outsourcing is based on the premise that the partner's employees will perform the tasks that have been outsourced in parallel with equivalent tasks for other customer organisations, and that standardisation across customers is what makes them efficient. The scope for the partner to contribute to the development of a specific business is thus limited. Another reason might be that the partner does not feel comfortable to allow an external party, such as the IT supplier, to work closely with the end customer, which is often required in order to develop operations in a relevant way. There may thus be good reasons to reject outsourcing, which we will now delve into further.

4.5.3. Reasons against Outsourcing

Cost of maintaining insight and monitoring. Letting an external actor manage parts of an organisation's operations presents a risk that the organisation will have less insight into how they are run than if operations were run by internal personnel. First, the organisation does not have the same physical opportunities for insight and monitoring when it is not on the same premises, or even in the same country (if the organisation has chosen an offshoring or captive solution). Secondly, buyers and suppliers may have different cultures and different approaches to communicating (which can of course also be the case between an internal unit within an organisation and the rest of the organisation), which can impede insight and monitoring. If the supplier is completely independent of the organisation, it also has its own incentives, which may not be fully aligned with those of the buyer, while an in-house department is likely to be more cognisant of, and concerned with, the organisation's goals and needs, for instance when the supplier must be chosen and the contract drafted, as well as in the ongoing relationship as progress is discussed and adjustments made. Travel, sometimes long-haul, is required when operations are outsourced abroad. Add to this any cultural conflicts and language barriers, and a relatively complex picture of the buyer/supplier relationship emerges.

Beijer Bygg, a Swedish chain of builders' merchants, has chosen not to outsource its IT to any significant extent, explaining that it is difficult to assure things like quality and development by means of a contract. By handling operations in-house, they believe they will be better equipped to ensure that they are sufficiently up-to-date with any development projects. For other, mainly large, organisations, it has instead been popular to try and create a version of the buyer/seller relationship internally, for example to ensure the quality of development and operations in IT through variations of the ITIL governance model. In such cases, the organisation prepares a service catalogue—a definition of the services to be delivered and specification of the parties responsible for ordering and executing each service—and reaches an agreement on the level of service delivery in scope and quality (SLA and OLA, service-level agreement, operational-level agreement). The value of proximity versus distance, and of informal dialogue versus formal contract drafting to establish the content of deliveries thus varies among organisations and, to an extent, over time. While some argue that quality-enhancing professionalisation requires formalisation, others argue that close and trusting dialogue is the foundation of quality and efficiency.

Loss of skills. One argument for retaining a sub-operation within the organisation is that there is otherwise a risk that the organisation will lose valuable skills. Even skilled buyers of services need to understand what they are trying to buy. First, it is not clear if someone who is only a buyer will be able to maintain that skill and secondly, it is not certain that skilled employees will stay with the organisation and transition to an exclusive buyer role. To be a skilled supplier, the supplier may need to understand the business or organisation to which it is delivering services. It may seem tempting to reduce the workforce through outsourcing, but the challenge is to retain a sufficient skills base. If, for example, an organisation shuts down the IT department and outsources its operations, what happens when employees leave? In an initial step, the organisation's employees may be transferred to the outsourcing partner and take their knowledge about the organisation and internal relationships with them, but as time passes, there is a risk that this knowledge will fade, and if the organisation switches suppliers, the training of the supplier will have to start from scratch.

As discussed earlier in this chapter and elsewhere, the IT department should cooperate closely with the rest of the organisation (if there is a separate IT department) to ensure that the use of IT will create value. If IT personnel exit an organisation, there is also a risk that valuable skills in the intersection between IT and operations will be lost. This has been argued in the banking industry, for instance, which has for a long time avoided outsourcing IT that supports its core business. This can become a problem in the event that an organisation no longer wants to work with its IT supplier. It might have been bought out and transformed into a supplier that no longer fits the needs of the organisation, and in that case, the organisation cannot quickly bring that part of its IT operations in-house. Any loss of skills can also constitute a problem even in a supplier relationship that is working well. As mentioned in the previous section, the relationship must be managed. What happens when things change externally and the organisation needs to review its value proposition? Should digitisation play a more prominent role in how the organisation competes? Or could there be better ways to use existing IT that the organisation should adopt? When essential parts of IT operations are contracted out to a supplier, there is a risk that the organisation will lack the skills necessary to respond rapidly to such changes.

Data exposure. Another fear sometimes expressed about outsourcing is that the organisation's data will be more exposed. This applies not least to cloud solutions, where data is processed in a public cloud via the Internet. In some contracts with public cloud suppliers, it is not even certain that the buyer will be given insight into how its own data is processed. As mentioned above, there are hopes that Swedish government agencies will be able to achieve significant savings by putting parts of their IT in the cloud, but the National Defence Radio Establishment of Sweden has recommended that the agencies join together and create a private cloud, specifically for data security. There are also legal requirements in the EU for how customer data must be handled from a security perspective. This affects where cloud service providers choose to locate their data centres. IBM works with local firms to create data centres that are situated in the country where the buyer wants to keep their data. The company also has its own networks, so that data traffic between data centres is free and more predictable than via the Internet's best-effort delivery. HL, a company that supplies retail stores with solutions for the placement and display of merchandise, manages its infrastructure and operations through the Amazon cloud. The servers are in Ireland and are thus covered by relevant EU legislation. Outsourcing outside the cloud can of course also involve risks. The agreement between the Swedish Transport Agency and IBM, mentioned in Section 4.5.2, has been intensively covered in the Swedish media. This is because of the emergence of information that sensitive data was possibly exposed to IBM employees who had not been authorised by the Swedish Transport Agency.

Another question that can arise is that of who owns the data circulating in the cloud, especially once a subscription has ended. In the light of these problems and risks, many organisations have doubts about the extensive use of cloud services. As we have previously discussed, however, it is highly possible that some organisations are already using them. In these contexts, there is discussion of "shadow IT" and "bring your own device", the phenomenon whereby employees choose their own IT tools, which the central organisation may not have approved or even know about. The pricing structure for cloud services means that they can be used without a big budget. To a certain extent, they can even be financed by other means, so that the extent of use is not associated with the price. Applications like Dropbox and Google Docs may generate commercial and legal risks if employees store business-critical data there and, for example, unauthorised individuals gain access to the documents. Services like Facebook can also put users in a subordinate position *vis-à-vis* the supplier, which decides what can be posted and can choose to impose new requirements on the user before providing continued access to existing data. This may be regarded as similar to the risk that, for example, service providers delivering outsourced customer service, who have learned an organisation's business and products and are in contact with its customers, could choose to end the relationship or impose new demands on the business in order to continue acting as an outsourcing partner for it.

The above reasons for and against outsourcing are often related to the ability to manage whatever product or service is outsourced. This topic will be discussed in the next section.

4.6. Management of Outsourcing

If an organisation chooses to outsource, the activities that are outsourced need to be managed so that they contribute to the goals of the organisation. Such management roughly concerns two things: first, procurement competence, the initial stage when a relationship is to be established, and second, the ongoing management when the relationship has been established.

4.6.1. Procurement Competence

As we noted in the previous section, it is important to understand what is purchased, so that price and quality are reasonable and there are favourable conditions for continued delivery during the intended period of outsourcing. If systems development is outsourced, there must not only be a well-founded idea of the desired functionality now and in the future; but also an ability to assess the extent to which the developed application can be adapted to the future needs of the organisation. Is the development taking place on a platform that several suppliers can handle, and that can be expected to last for the foreseeable future? Do systems architecture and coding allow for future adjustments, or will such adjustments be very complicated to implement? To some extent, assessments on these questions can be made by third parties, but a certain degree of insight on the area is necessary in order to be able to determine whether such an assessment is credible and competently carried out. The less that a buyer knows and understands, the greater their risk of becoming completely dependent on a counterpart whose competence cannot be judged.

Procurement competence is a difficult area, as the basis for deciding to outsource activities to external suppliers is often the notion that your own organisation should devote its resources to something else. This can make it difficult both to maintain procurement competence within the organisation, and to persuade those employees who possess important insights to participate in the procurement process. Recommendations usually include gathering a wide range of people in procurement projects. There should be people with a management perspective on what is to be purchased, as well as people with practical experience of the activities that are to be affected by the procurement. If the area is of strategic importance, people from the management team should be more than marginally involved. Furthermore, those involved should have a standing in the organisation that the other parties concerned will accept and in whose judgments and decisions they will have confidence. A proper market survey should be conducted, with potential suppliers roughly sorted according to relevant criteria, and two or more thoroughly evaluated in order to assess their ability to deliver the desired services both in the short term and in the longer term. If possible, the evaluation process should also follow a standardised model. See Chapters 5 and 6 on structured decisions and procurement.

Of course, such a recommendation is sound from the perspective of making as well-grounded a decision as possible, but if there are competent and knowledgeable people within the organisation, they probably have other tasks on which they should spend their time and which they, or their managers, see as more of a priority. Recommendations generally advocate long-term, strategic thinking. In both organisational and private life, it is easy to find examples of how more long-term considerations are sidelined in favour of short-term issues. Applying these recommendations in practice is therefore difficult. The better the procurement, the more favourable the conditions for the outsourcing to work well. The less insightful the procurement process, the more important it is to follow up and try to control the supply of resources or services. We now turn our attention to the continuous monitoring and management of outsourcing.

4.6.2. Continuous Monitoring and Management

Few or many suppliers? One question is how many suppliers are required. Having few strategic partners can provide opportunities for long-term cooperation and mutual adjustment. At the same time, it can make it difficult to judge whether the partner is competitive, and it becomes problematic to change suppliers. Having many alternative suppliers makes comparability easier, but in that position no counterpart will see any reason to try to adapt long-term, as there is too great a risk that they will be replaced or will not receive large order volumes. The general view of a balance between a few close collaborations and several at arm's length varies. Trends are not unambiguous and ideals are not consistent with practice. The CIO of the Northern European insurance firm IF reports that in the past decade they have gone from one to nine suppliers, primarily in infrastructure. They did this to obtain better prices and to increase their skills base. On the other hand, some would argue that there is a trend, among both private and public organisations, to reduce the number of suppliers so as to gain a better overview and to avoid escalating costs. In recent years, the Swedish Social Insurance Agency has reduced its IT suppliers from over forty to three.

Short-term or long-term contracts? This is a question that is in part related to the issue above: the length of a contract with a supplier. Ten-year contracts were once not uncommon, but nowadays outsourcing contracts are typically shorter. More long-term contracts seem to increase the risk of the supplier not maintaining a good level of service. Long-term contracts can also "lock in" the customer, so that the changing needs that arise as an organisation develops and technology changes cannot be met within the scope of the existing agreement. In short-term contracts, on the other hand, the responsibility for development relies largely on market-driven competition; a supplier with only a short-term contract will find it difficult to motivate long-term investments to adapt to the buyer. Long-term contracts make it more reasonable for the organisation to try to develop alongside the counterpart, and may therefore be important for organisations resorting to outsourcing to transform the product or service that is being outsourced. Having said that, it is wise to incorporate some amount of flexibility in a long-term contract.

Hard or soft governance? As discussed earlier in the chapter, there are different forms of outsourcing, and outsourcing may involve different aspects of IT-related operations. The governance mechanisms that are most suitable therefore depend on the situation. A study by Jérôme Barthélemy shows that outsourcing handled through detailed contracts tends to be of the simpler kind, such as a data centre, where the supplier has the potential for economies of scale. Outsourcing that is governed by such contracts also tends to generate cost efficiency, but not other types of benefits. In contrast, outsourcing relationships where governance is based on trust rather than strict formalisation often include complex projects, such as development. The use of soft, trust-based governance tends to be related to benefits such as goal attainment. These patterns are in line with the reasoning in Section 4.2 that the suitability of hard and soft governance depends on how dynamic an organisation's environment is, however the Barthélemy study is not sufficient to establish causal relationships between governance and outcomes.

Complex exchanges are difficult to formulate in a detailed contract; it is difficult to anticipate everything that will happen in the relationship, and questions about what should and should not be included can more easily arise. In such a relationship, the softer side of governance becomes important. The combination of both detailed contracts and softer governance seems advantageous, as it has been shown to provide both cost efficiency and goal attainment. It is used in both simpler and more complex outsourcing arrangements. For example, it is possible to specify in a contract that "new technology" is to be used and that a certain amount of money should be spent on technology development every year. It is difficult to be more detailed than this when it comes to technology development, and this is where soft, trust-based governance plays an important role. According to Deloitte's 2018 global outsourcing survey, purchasing organisations are also increasingly incorporating incentives for innovation in their outsourcing contracts, not only in the form of compensation, but also by moving additional services to the supplier as they innovate. As mentioned earlier, many IT buyers now want their supplier not only to deliver something predetermined but also to deliver improvements and innovations. Studies of the procurement of innovations in general indicate that there are risks to being overly detailed as a buyer, because new thinking can be stifled.

Contracts can be useful in the beginning, so that trust can develop over time, and in the same way, a contract that is not fulfilled will probably result in a lack of trust. There have been some studies on buyer-supplier relationships that were initially governed by detailed contracts. These have suggested that only when the contract management is abandoned and a person responsible for the relationship with each party is introduced, do buyers and suppliers get to know each other better and more clearly understand how they might mutually gain from the relationship. For example, loosely formulated incentives were created for possible development and innovations implemented by the supplier. One conclusion is that even fairly simple exchanges need relationship management to some extent.

As previously mentioned, it is unlikely that a seemingly standardised service can easily be purchased and integrated like a Lego brick into our own operations without any maintenance. Outsourcing tends to be a learning journey, sometimes termed an "experience good"; what defines quality and value for money only becomes evident during the journey, and is difficult to determine before the contract is signed. It is therefore important to continually follow up and try to get an idea of how well the outsourced activities are functioning. If users of the services are internal to the organisation, it is possible to evaluate, although obviously not easy. If users are external, for example, if the outsourced activities relate to customer service or a webpage, the contract may need to be supplemented with specific types of feedback required from customers or the supplier's interactions with them. Can they (easily) find what they are looking for? Are their questions answered adequately? Do they perceive the supplier's services as part of the business, or does it appear to be a separate, independent element? Contracts can be comprehensive, however. The supplier and the staff working on behalf of the organisation need to understand the role of such feedback in operations if the business is to maintain good quality. Just as in internally managed activities, trust-based collaboration is necessary, as well as insight into how the activities fit in with the rest of the organisation.

Ongoing collaboration and operations may be full of small decisions, but these will to a large extent be guided by the routines and habits that are developing gradually over time. The standardisation of routines is often considered as something leading to good work outcomes, but this is far from certain. Only certain tasks are so naturally standardisable that it makes sense to use routines to produce quality-assured results. Often, the performers' insights on what constitutes a successful result are what enable them to utilise their judgment and opportunities for feedback so as to deliver good quality. It is no coincidence that agile methods have become popular in systems development; fundamental element of agile methods is the close dialogue between supplier and customer. This close dialogue allows for prioritising and re-prioritising future development in the light of current insights into successes and challenges. In more complex types of outsourcing and in new outsourcing relationships, such close collaboration between clients and suppliers is extremely valuable.

4.7. Chapter Summary

In this chapter, we have described different ways of allocating work and responsibilities related to digitisation. A clear trend is that employees from many parts of an organisation are taking increased responsibility for questions related to digitisation. This is in line with the idea that digitisation is no longer just a technical issue, but above all an organisational issue. Depending on the size and type of strategy of the organisation, the responsibility and involvement of operations—and IT specialists—may be either at the management level or at the local level. Employee involvement can include everything from discovering new ways to digitise operations to effectively managing established IT use. The view on how digitisation can strategically contribute to operations will probably vary depending on where in the organisation someone is located, so coordination between different employees should be characterised by, for example, shared knowledge, positive experiences of previous joint projects, and regular communication without overly specialised jargon. In this chapter, we have also presented various ways of allocating an organisation's digitisation to external actors, the advantages and disadvantages associated with doing so, and how such buyer-seller relationships can be governed in terms of, for example, a balance between detailed contracts and a more flexible, trust-based attitude.

This chapter has dealt with the organisational aspects of digitisation, and the next chapter will focus more on how specific digitisation decisions can be handled. Regardless of who is involved in the organisation's digitisation and in what forms, decisions will have to be made, for example when choosing a supplier for outsourcing or evaluating whether to invest in a project. One part of such decision-making is the weighting of different preferences; preferences that may differ depending on who is involved in the decision. We will therefore now shift focus from the issues of roles, competencies, and skills, and how these relate to each other, to how decisions can be structured according to processes, criteria, and the weighting of criteria. This also entails a shift from what can be considered feasible, and even from current practices, to a more ideal world that many organisations are still nowhere near.

4.8. Reading Tips

Below are two examples of large digitisation projects with differing governance approaches briefly discussed in this chapter. The first is a drawn-out, rather painful project, in which ambitions regarding business aspects are gradually diminished in order to get a technically functioning system in place. The second is a highly business-focused project, where the learnings from the first contribute to a different governance approach, and to a digitisation effort where organisational and technical development are more in line with one another.

- Westelius, Alf (2006). Muddling through: The life of a multinational, strategic enterprise systems venture at BT Industries. *Linköping Electronic Articles in Computer and Information Science* 10 (1), 46 pages, http://liu.diva-portal.org/smash/get/diva2:256674/FULLTEXT01.pdf.
- Valiente, Pablo and Westelius, Alf (2007). Sustainable Value of Wireless ICT in Communication with Mobile Employees. In: Per Andersson, Ulf Essler and Bertil Thorngren (eds), *Beyond Mobility*. Lund/Stockholm: Studentlitteratur/EFI, pp. 175–206.

For broad empirical studies on the interaction between IT and operations, see Weill's article and Andriole's more recent article. A more in-depth study of the relationship between IT and operations is found in Jansson's article on e-government in municipalities. Practitioner-oriented journals, such as *CIO,* have also presented surveys on the theme.

- Weill, Peter (2004). Don't just lead, govern: How top-performing firms govern IT. *MIS Quarterly Executive* 3 (1), pp. 1–17.
- Andriole, Stephen J. (2015). Who owns IT? *Communications of the ACM* 58 (3), pp. 50–57, http://dx.doi.org/10.1145/2660765.
- Jansson, Gabriella (2013). Politikens betydelse för e-förvaltning: Om lokala IT-projekt och det demokratiska ansvarsutkrävandet [The importance of politics for e-governance: On local IT projects and democratic accountability, in Swedish]. *Scandinavian Journal of Public Administration* 17 (2), pp. 103–125, https://ojs.ub.gu.se/index.php/sjpa/article/view/2480/2210.

- CIO Sweden (2016). En ny, digitalare governance. [A new, more digital governance, in Swedish]. 5, pp. 16–19.

For those who want to know more about the balance between centralised and decentralised governance in general, and how it can be adapted to an organisation's strategic focus, we recommend the following anthology, which contains examples from many industries.

- Jannesson, Erik; Nilsson, Fredrik and Rapp, Birger (eds) (2014). *Strategy, Control and Competitive Advantage: Case Study Evidence.* Berlin/Heidelberg: Springer, https://doi.org/10.1007/978-3-642-39134-7.

A nice overview of the development of the CIO role and its different possible foci in terms of innovation and management is given in Chun's and Mooney's article. A paper by Magnusson and colleagues provides more recent insight into the role in the Swedish context.

- Chun, Mark and Mooney, John (2009). CIO roles and responsibilities: 25 years of evolution and change. *Information & Management* 46 (6), pp. 323–334, https://doi.org/10.1016/j.im.2009.05.005.
- Magnusson, Johan; Högberg, Erik and Sjöman, Hampus (2019). How the West was lost: Chief Information Officers and the battle of jurisdictional control. In: *Proceedings of the 52nd Hawaii International Conference on Information Science*, Grand Wailea, Maui, January 8-11, 2019, https://doi.org/10.24251/hicss.2019.746.

For a more general understanding of what it means to combine innovation and novel thinking on the one hand, and more repetitive, maintenance-related tasks on the other hand, and how such a combination of tasks can be supported within an organisation, the following article describes the phenomenon of ambidexterity.

- Adler, Paul S.; Goldoftas, Barbara and Levine, David I. (1999). Flexibility versus efficiency? A case study of model changeovers in the Toyota production system. *Organization Science* 10 (1), pp. 43–68, https://doi.org/10.1287/orsc.10.1.43.

The phenomenon has been studied from many perspectives, however, and the journal *Organization Science* devoted an entire issue to it: 2009, 20(4).

If you want to know more about the social dimension of the relationship between IT and operations managers, or how the mutual understanding and knowledge between these parties can be deepened, Reich and Benbasat's study of Canadian insurance companies is recommended, as well as Bensaou and Earl's study in the Japanese context.

- Reich, Blaize H. and Benbasat, Izak (2003). Measuring the information systems business strategy relationship: Factors that influence the social dimension or alignment between business and information technology objectives. In: Galliers, Robert D. and Leidner, Dorothy E. (eds), *Strategic Information Management: Challenges and Strategies in Managing Information Systems*. Rochester: Butterworth-Heinemann, pp. 265–310, https://doi.org/10.4324/9780080481135-19.
- Bensaou, Ben M. and Earl, Michael (1998). The right mind-set for managing information technology. *Harvard Business Review* 76 (5), pp. 119–128.

For a more general understanding of the challenges of integrating different parts of an organisation, we recommend the following two seminal articles.

- Daft, Richard and Lengel, Robert (1986). Organizational information requirements, media richness and structural design. *Management Science* 32 (5), pp. 554–571, https://doi.org/10.1287/mnsc.32.5.554.
- Galbraith, Jay R. (1974). Organization design: An information processing view. *Interfaces* 4 (3), pp. 28–36, https://doi.org/10.1287/inte.4.3.28.

If you want to read more about IT outsourcing, underlying motives and how outsourcing relationships can be governed, we recommend Blaskovich and Mintchik's review article, as well as Barthélemy's and Kern et al.'s empirical studies on how different control mechanisms function in different outsourcing situations. *CIO* and various consulting firms have also published surveys and investigations on IT outsourcing.

- Blaskovich, Jennifer and Mintchik, Natalia (2011). Information technology outsourcing: A taxonomy of prior studies and directions for future research. *Journal of Information Systems* 25 (1), pp. 1–36, https://doi.org/10.2308/jis.2011.25.1.1.
- Barthélemy, Jérôme (2003). The hard and soft sides of IT outsourcing management. *European Management Journal* 21 (5), pp. 539–548, https://doi.org/10.1016/s0263-2373(03)00103-8.
- Kern, Thomas; Kreijger, Jeroen and Willcocks, Leslie (2002). Exploring ASP as sourcing strategy: Theoretical perspectives, propositions for practice. *Journal of Strategic Information Systems* 11 (2), pp. 153–177, https://doi.org/10.1016/s0963-8687(02)00004-5.
- Whitelane Research and PA Consulting Group (2016). *IT Outsourcing Study 2016 Nordics*. Report published in collaboration between Whitelane Research and PA Consulting Group.
- CIO Sweden (2015). Trenden går mot multimiljö: Så IT-sourcar svenska CIOer 2015 [The trend is moving towards multi-environments: How Swedish CIOs source IT 2015, in Swedish]. http://cio.idg.se/2.1782/1.630340/sa-sourcar-svenska-cio-er-2015.

Standardised routines have many advocates, but this seminal article illustrates how such routines may conflict with professional knowledge and development:

- Brown, John S. and Duguid, Paul (1991). Organizational learning and communities-of-Practice: Toward a unified view of working, learning, and innovation. *Organization Science* 2 (1), Special issue: Organizational learning: Papers in honour of (and by) James G. March, pp. 40–57, https://doi.org/10.1287/orsc.2.1.40.